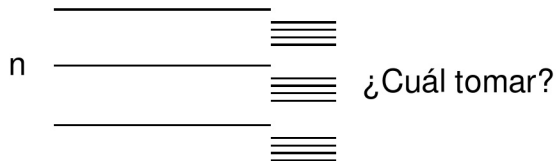


### Caso Degenerado:

Sea  $\{\phi_n^{(i)}\}$  un conjunto finito de estados con la misma energía.

$$(\phi_m^j, \phi_n^i) = \delta_{m,n} \delta_{j,i}$$



$$\psi_n = N(\lambda) \left\{ \sum_i \alpha_i \phi_n^{(1)} + \lambda \sum_{k \neq n} C_{nk}^{(1)} \sum_i \beta_i \phi_k^{(i)} + \dots \right\}$$

Debemos encontrar  $\alpha_i, \beta_i$

$$(H_0 + \lambda H_1) \psi_n = E_n \psi_n$$

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \dots$$

$$C_{nk}(\lambda) = \lambda C_{nk}^{(1)} + \lambda^2 C_{nk}^{(2)} + \dots$$

Reemplazando  $\psi_n$  en la ecuación de Sch. obtenemos para los términos proporcionales a  $\lambda$ :

$$\begin{aligned}
 & H_0 \sum_{k \neq n} C_{nk}^{(1)} \sum_i \beta_i \phi_k^{(i)} + H_1 \sum_i \alpha_i \phi_n^{(i)} \\
 &= E_n^{(1)} \sum_i \alpha_i \phi_n^{(i)} + E_n^{(0)} \sum_{k \neq n} C_{nk}^{(1)} \sum_i \beta_i \phi_k^{(i)}
 \end{aligned}$$

$$(\phi_n^j, \phi_k^i) = 0 \quad n \neq k$$

$$(\phi_n^{(j)}, \phi_n^{(i)}):$$

$$\sum_i \alpha_i (\phi_n^j, H_1 \phi_n^i) = E_n^{(1)} \sum_i \alpha_i \underbrace{(\phi_n^j, \phi_n^i)}_{\delta_{m n j i}}$$

$$E_n^{(1)} \alpha_j = \sum_i \alpha_i (\phi_n^j, H_1 \phi_n^i)$$

Corrección en 1° orden para la energía.

Consideremos dos estados con la misma energía:

$$E_n^{(1)} \alpha_j = \sum_i \alpha_i (\phi_n^j, H_1 \phi_n^i)$$

$$E_n^{(1)} \alpha_1 = \alpha_1 (\phi_n^1, H_1 \phi_n^1) + \alpha_2 (\phi_n^1, H_1 \phi_n^2)$$

$$E_n^{(1)} \alpha_2 = \alpha_1 (\phi_n^2, H_1 \phi_n^1) + \alpha_2 (\phi_n^2, H_1 \phi_n^2)$$

Sea  $h_{ji} = (\phi_n^j, H_1 \phi_n^i)$

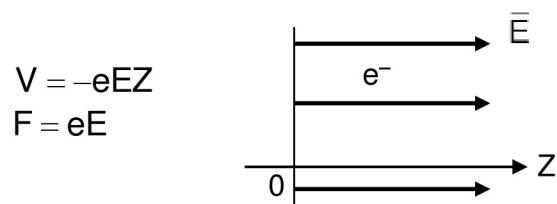
$$\begin{cases} h_{11} \alpha_1 + h_{12} \alpha_2 = E_n^{(1)} \alpha_1 \\ h_{21} \alpha_1 + h_{22} \alpha_2 = E_n^{(1)} \alpha_2 \end{cases} \quad \begin{matrix} (1) \\ (2) \end{matrix}$$

$$\begin{pmatrix} h_{11} & h_{12} \\ h_{21} & h_{22} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix} = E_n^{(1)} \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$$

$$\sum |\alpha_i|^2 = 1$$

Ejemplo:

Átomo en un campo eléctrico uniforme ( $\vec{E}$ ):



$n=1 \quad \ell=0 \quad m=0$

$\psi_{100}$  no es degenerado

$n=2 \quad \ell=0, 1$   
 $m=0 \downarrow m=-1, 0, 1$   
 $|n \ell m\rangle: |2, 1, 1\rangle, |2, 1, 0\rangle, |2, 1, -1\rangle$   
 $|2, 0, 0\rangle$

$\eta = 3$

$\ell = 0, 1, 2$   
 $m = 0 \downarrow -2, -1, 0, 1, 2$   
 $-1, 0, 1$

$$h_{ji} = (\phi_n^j, H_1 \phi_n^i)$$

Se trata de encontrar la matriz de V:

$$h_{12} = \langle 200 | -eEZ | 210 \rangle$$

$$= -eE \langle 200 | Z | 210 \rangle$$

$$= -eE \int \psi_{200}^* Z \psi_{210} d^3\vec{r}$$

$$V = \begin{array}{c|cccc} & |200\rangle & |210\rangle & |211\rangle & |21-1\rangle \\ \hline |200\rangle & 0 & -3ea_0E & 0 & 0 \\ |210\rangle & -3ea_0E & 0 & 0 & 0 \\ |211\rangle & 0 & 0 & 0 & 0 \\ |21-1\rangle & 0 & 0 & 0 & 0 \end{array}$$

$$a_0 = 0,52 \text{ \AA} : \text{Radio de Bohr}$$

$$h_{12} = -eE \int_0^\infty \int_0^\pi \int_0^{2\pi} \psi_{200}^* r \cos\theta \psi_{210} r^2 \sin\theta dr d\theta d\phi$$

$$O^{(2)}, \underbrace{\pm 3ea_0E}_{\Delta \text{ o valores}}$$

$$\begin{array}{l} 2s, 2p \left\{ \begin{array}{l} \frac{1}{\sqrt{2}} (|2,1,0\rangle + |2,0,0\rangle) \\ |2,1,1\rangle \\ |2,1,-1\rangle \\ \frac{1}{\sqrt{2}} (|2,1,0\rangle - |2,0,0\rangle) \end{array} \right. \end{array} \quad \begin{array}{l} E_2 = E_2^0 + 3ea_0E \\ \\ \\ E_2 = E_2^0 - 3ea_0E \end{array}$$

$$\begin{bmatrix} 0 & -\Delta E \\ -\Delta E & 0 \end{bmatrix} \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} = \pm \Delta E \begin{bmatrix} \alpha_1 \\ \alpha_2 \end{bmatrix} \Rightarrow \begin{array}{l} \alpha_2 = \alpha_1 \\ -\alpha_1 = \alpha_2 \end{array}$$

$$\Delta E = 3ea_0E$$